

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Eighth Semester of B. Tech. (CL) Examination

May 2014

CL 431 Earthquake Engineering

Date: 09.05.2014, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.
5. IS 1893: 2002 Part-I, IS 4326: 1993, IS 13920: 1993, IS 13827:1993, & IS 13828: 1993 are permitted.

SECTION - I

Q - 1 Answers the following as per IS 13920:1993:

[07]

1. What should be width to depth ratio of the flexural member?
2. What should be minimum grade of concrete for all buildings?
3. The redistribution of moments shall be used only for vertical load moments and not for lateral load moment. True or False?
4. What should be minimum diameter of the bar forming a hoop?
5. Give IS consideration for hoop spacing.
6. In which conditions lap splices should not be provided in a beam.
7. What should be factored axial stress on the member due to earthquake loading?

Q - 2 (a) A singly reinforced beam having width $b = 300$ mm and depth $D = 600$ mm is reinforced with 3 nos of 20 mm diameter bars. Check whether this beam satisfied the criteria for reinforcement as per IS 13920:1993.

[02]

(b) Attempt any Two:

[12]

1. A free vibration test is conducted on an empty elevated water tank. A cable attached to that tank applies a lateral force of 73 kN force and pulls the tank horizontally by 5.1 cm. The cable is suddenly cut and the resulting free vibration is recorded. At the end of four complete cycles, the time is 2.0 sec and the amplitude is 2.5 cm. From these data compute the following: (i) damping ratio, (ii) natural period of undamped vibration, (iii) stiffness, (iv) weight, (v) damping coefficient and (vi) number of cycles required for the displacement amplitude to decrease to 0.5 cm.
2. Determine the natural frequency of the system shown in Figure 1 consisting of 23 kg mass attached to a horizontal cantilever beam through coil spring k . The

cantilever beam has a thickness $t = 6.35$ mm, a width $b = 25.4$ mm, Modulus of elasticity $E = 2 \times 10^5$ N/mm², and a length $L = 317.5$ mm. The coil stiffness $k = 1949$ N/m.

3. A sensitive instrument with mass of 45.45 kg is to be installed at a location where the vertical acceleration is 0.1 g at a frequency of 10 Hz. The instrument is mounted on a rubber pad of stiffness 14044 N/m, damping is such that ratio of the system is 10%. Determine (i) what acceleration is transmitted to the instrument? (ii) if the instrument can tolerate an acceleration of 0.005 g, suggest a suitable solution assuming that same rubber pad is used.

Q – 3 (a) Give criteria as per IS 13920:1993 for

[02]

- (i) Length of special confining reinforcement
- (ii) Maximum spacing of hoops using special confining reinforcement

(b) Attempt any One:

[12]

1. For the residential RCC (SMRF) building shown in **Figure 2** founded on soft soil and situated in zone V shown in figure. Compute the design lateral force at each floor in each mode using dynamic analysis procedure. Given the free vibration analysis results as follows,

Frequency $\{\omega\} = \{47.832, 120.155, 167.00\}$ rad/sec

Modes: $\{\phi_1\} = \{1.0, 0.759, 0.336\}$

$\{\phi_2\} = \{1.0, -0.805, -1.157\}$

$\{\phi_3\} = \{1.0, -2.427, 0.075\}$

Seismic Weights: $W_1 = W_2 = W_3 = 1962$ kN

Stiffness $k_1 = k_2 = 160 \times 10^3$ kN/m and $k_3 = 240 \times 10^3$ kN/m

2. A three storied building has lumped floor weights from bottom to top as 688 kN, 688 kN & 640 kN with storey stiffness of each floor 47524.5 kN/m. Perform the free vibration analysis & determine natural frequencies and corresponding mode shape coefficients. Also sketch the mode shapes.

SECTION – II

Q – 4 (a) Attempt the following:

1. According to IS 1893: 2002 (Part 1), the ratio (I/R) shall not be greater than 01
2. Strong beam-weak column concept is preferred in earthquakes resistant design. True or False? Justify your answer. 02
3. The engineers do not attempt to make earthquake proof structures. Justify this 02

statement.

4. Enlist the major tectonic plates of Earth's surface.

02

Q - 5 Attempt any Two:

[14]

- (a) For a four storeyed RCC office building located in zone V and resting on hard rock, compute the seismic forces as per IS-1893-2002 equivalent static procedure. Height of first is 4.2 m and the remaining three stories are of height 3.2 m each. Plan dimensions (length and width) of the structure are 15 m x 20 m. The moment resisting frame is without infill walls. Dead load on floor 12 kN/m² on floors and 10 kN/m² on roof. Live = 4 kN/m² on floors and 1.5 kN/m² on roof.

- (b) Calculate the torsional shear forces in a one-storey shear wall masonry structure with a rigid diaphragm roof for the following data. There are four shear walls, with relative rigidity of each wall is as shown in **Figure 3**.

Center of mass: $x_{cm} = 8.69$ m from wall-II, $y_{cm} = 3.19$ m from wall-IV

Center of rigidity: $x_{cr} = 4.32$ m from wall-II, $y_{cr} = 1.20$ m from wall-IV

Height of parapet wall = 1 m

Height of wall upto roof levels = 3 m

Seismic zone = V, $Z = 0.36$, $I = 1.0$, $R = 1.5$; $S_a/g = 2.5$

Self weight of the roof = 3.0 kN/m²

- (c) 1. Irregularities of mass, stiffness, and strength are not desirable in buildings situated in earthquake prone areas. Describe using diagrams how these occur and affect the building.
2. Define response spectra and discuss its significance in analysis of structure.

Q - 6 Attempt any Two:

[14]

- (a) Determine the rigidity of the shear wall as shown in **Figure 4**, in terms of Et.
- (b) The masonry building as shown in **Figure 5** is made up of North, South, East and West walls. The rigidity of all four walls is 0.462, 0.538, 0.5 and 0.5 respectively. Height of parapet wall = 1 m, height of walls up to roof levels = 3 m, self weight of roof = 3.0 kN/m², self weight of the wall = 5.0 kN/m². Calculate location of centre of mass and center of rigidity.
- (c) Define bands. At what levels in a masonry building would you provide them? Give Justifications for each of them.

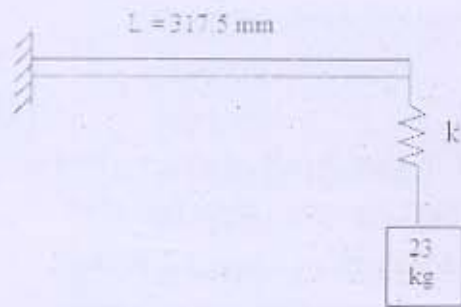


Figure 1

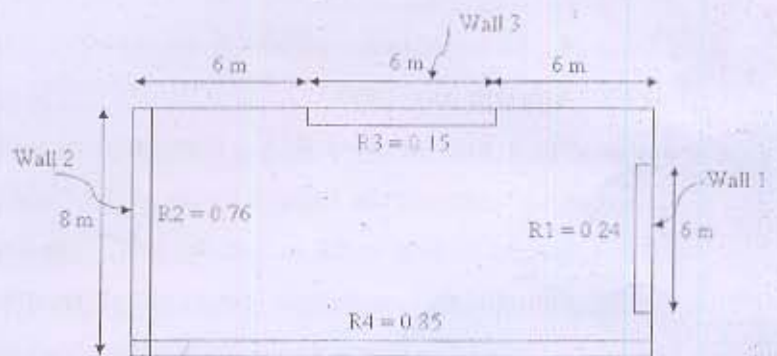


Figure 3

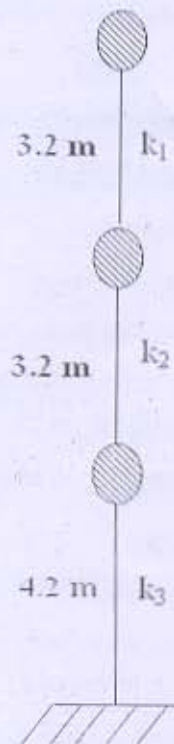


Figure 2

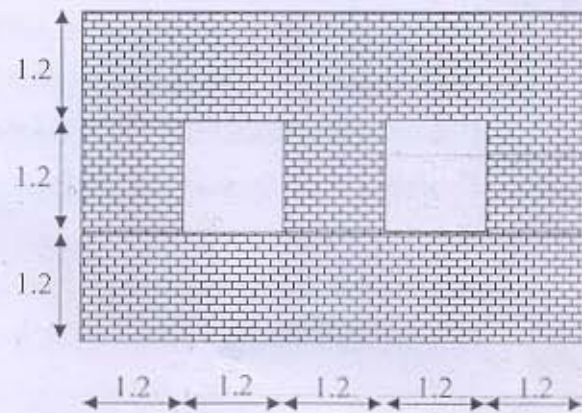


Figure 4 (All dimensions are in meter)

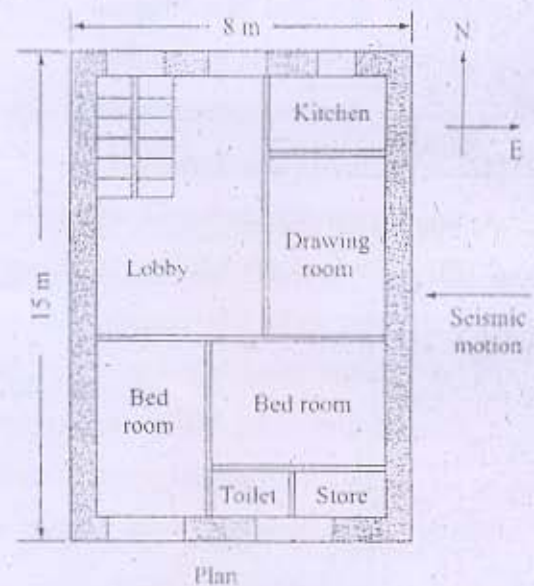


Figure 5
